Installation and Setup

Java (JDK) Installation:

Step 1: Type the command **sudo apt install openjdk-11-jdk -y in the terminal**

It will install the OpenJDK 17 (Java Development Kit) and accepts installation automatically.

```
vboxuser@prathwish03:~$ sudo apt install openjdk-11-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.30+7-1ubuntu1~22.04).
The following packages were automatically installed and are no longer required:
  linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic
  linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vboxuser@prathwish03:~$
```

The installation will start like this.

Step 2: Installation can be confirmed by running the command **java -version**

```
vboxuser@prathwish03:~$ java -version
openjdk version "11.0.30" 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode,
 sharing)
vboxuser@prathwish03:~$
```

Maven Installation:

Step 1: Type the command **sudo apt install maven** in the terminal. It will install Apache Maven from Ubuntu's package repository.

```
vboxuser@prathwish03:~$ sudo apt install maven
[sudo] password for vboxuser:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vboxuser@prathwish03:~$
```

Step 2: Installation can be confirmed by running the command **mvn -version**

```
vboxuser@prathwish03:~$ mvn -version
Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 11.0.30, vendor: Ubuntu, runtime: /usr/lib/jvm/java-11-openjdk-amd
64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-101-generic", arch: "amd64", family: "unix"
vboxuser@prathwish03:~$
```

Gradle Installation:

Step 1: Type the command **sudo apt install gradle** in the terminal

It installs Gradle build automation tool from Ubuntu's package repository.

```
vboxuser@prathwish03:~$ sudo apt install gradle
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gradle is already the newest version (4.4.1-13).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vboxuser@prathwish03:~$
```

Step 2: To check the proper installation of Gradle run the command **gradle –version**

```
vboxuser@prathwish03:~$ gradle --version
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.reflection.CachedClass
 (file:/usr/share/java/groovy-all.jar) to method java.lang.Object.finalize()
WARNING: Please consider reporting this to the maintainers of org.codehaus.groov
y.reflection.CachedClass
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflect
ive access operations
WARNING: All illegal access operations will be denied in a future release

------------------------------------------------------------
Gradle 4.4.1
------------------------------------------------------------

Build time:   2012-12-21 00:00:00 UTC
Revision:     none

Groovy:       2.4.21
Ant:          Apache Ant(TM) version 1.10.12 compiled on January 17 1970
JVM:          11.0.30 (Ubuntu 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OS:           Linux 6.8.0-101-generic amd64
```